

Docker Container Naming, Automatic Removal, and Image Versioning

To Begin with the Lab

Summary of the Lab

This lab demonstrated how to assign custom names to Docker containers, automatically remove containers after they exit using the `--rm` option, create versioned Docker images with tags, and deploy different application versions for effective container and image management.

- You can see that docker itself assign the names for containers but we can assign the custom name to a container instead of using the automatically generated name.

```
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-45-84:~$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED      STATUS      PORTS     NAMES
030e114d0a9b   static-app "httpd-foreground"   31 minutes ago   Exited (0) 25 minutes ago   80/tcp    nostalgic_nobel
8a6b8279ffdf   ubuntu    "/bin/bash"          3 hours ago     Exited (137) 33 minutes ago   hopefull_dirac
42b8a28dc825   ubuntu    "/bin/bash"          3 hours ago     Exited (0) 3 hours ago       pedantic_greider
f580899d95b6   ubuntu    "/bin/bash"          3 hours ago     Exited (0) 3 hours ago       gifted_brahmagupta
9791cea36102   ubuntu    "/bin/bash"          3 hours ago     Exited (0) 3 hours ago       youthful_sinoussi
f7552e7519a2   nginx     "/docker-entrypoint..." 3 hours ago     Exited (0) 3 hours ago       zealous_khayyam
2750e4b5b98f   nginx     "/docker-entrypoint..." 3 hours ago     Exited (0) 3 hours ago       practical_greider
7c179fd50eb7   nginx     "/docker-entrypoint..." 21 hours ago    Exited (137) 20 hours ago    wonderful_wiles
ubuntu@ip-172-31-45-84:~$
```

- Let's do this, run the command
- You will see Container ID, Image, Status, Name (static-app-instance)

```
sudo docker run -d -p 80:80 --name static-app-instance static-app
```

```
ubuntu@ip-172-31-45-84:~$ sudo docker run -d -p 80:80 --name static-app-instance static-app
557fa48e8a2f7effd3180c1ab32e6631c47c9ce506f874d618c6f44ec0d5b30b
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED      STATUS      PORTS     NAMES
557fa48e8a2f   static-app "httpd-foreground"   6 seconds ago   Up 5 seconds   0.0.0.0:80->80/tcp, [::]:80->80/tcp   static-app-instance
ubuntu@ip-172-31-45-84:~$
```

- Stop this container

```
sudo docker stop static-app-instance
```

- You can automatically remove a container after it exits.

```
sudo docker run --rm -d -p 80:80 --name static-app-instance1 static-app
```

- you can see with the same name we don't have the existing container

```
ubuntu@ip-172-31-45-84:~$ sudo docker run --rm -d -p 80:80 --name static-app-instance1 static-app
80ca89a1154cd8170c762f4485e89862c7da41a6c1f0348b5e/ccf966d853
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED      STATUS      PORTS     NAMES
80ca89a1154   static-app "httpd-foreground"   8 seconds ago   Up 7 seconds   0.0.0.0:80->80/tcp, [::]:80->80/tcp   static-app-instance1
ubuntu@ip-172-31-45-84:~$ sudo docker stop static-app-instance1
static-app-instance1
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-45-84:~$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED      STATUS      PORTS     NAMES
030e114d0a9b   static-app "httpd-foreground"   44 minutes ago   Exited (0) 39 minutes ago   80/tcp    nostalgic_nobel
8a6b8279ffdf   ubuntu    "/bin/bash"          3 hours ago     Exited (137) 46 minutes ago   hopefull_dirac
42b8a28dc825   ubuntu    "/bin/bash"          3 hours ago     Exited (0) 3 hours ago       pedantic_greider
f580899d95b6   ubuntu    "/bin/bash"          3 hours ago     Exited (0) 3 hours ago       gifted_brahmagupta
9791cea36102   ubuntu    "/bin/bash"          3 hours ago     Exited (0) 3 hours ago       youthful_sinoussi
f7552e7519a2   nginx     "/docker-entrypoint..." 3 hours ago     Exited (0) 3 hours ago       zealous_khayyam
2750e4b5b98f   nginx     "/docker-entrypoint..." 4 hours ago     Exited (0) 3 hours ago       practical_greider
7c179fd50eb7   nginx     "/docker-entrypoint..." 21 hours ago    Exited (137) 20 hours ago    wonderful_wiles
ubuntu@ip-172-31-45-84:~$
```

- Docker Image Tags

- Objective: Understand image versioning.
- Build Version 2
- Modify your index.html.
- Then build another image.

```
sudo docker build -t static-app:v2 .
```

- Verify:

```
sudo docker images
```

```
ubuntu@ip-172-31-45-84:~/app$ sudo nano index.html
ubuntu@ip-172-31-45-84:~/app$ sudo docker build -t static-app:v2 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  5.632kB
Step 1/2 : FROM httpd
--> 393435ee1a31
Step 2/2 : COPY . /usr/local/apache2/htdocs
--> d48b367577a2
Successfully built d48b367577a2
Successfully tagged static-app:v2
ubuntu@ip-172-31-45-84:~/app$ sudo docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
httpd:latest	393435ee1a31	177MB	47.6MB	
nginx:latest	ec4ed8b5299e	241MB	66MB	U
static-app:latest	697da8395c47	175MB	45.2MB	U
static-app:v2	d48b367577a2	175MB	45.2MB	
ubuntu:latest	b7f48194d4d8	160MB	45.3MB	U

```
ubuntu@ip-172-31-45-84:~/app$
```

- Run the New Version

```
sudo docker run -d -p 80:80 static-app:v2
```

```
ubuntu@ip-172-31-45-84:~/app$ sudo docker run -d -p 80:80 static-app:v2
f52c13ef739bcf87c19816541673bb8b4e61a3ffa7d983fe6e31c106884808a6
ubuntu@ip-172-31-45-84:~/app$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
f52c13ef739b	static-app:v2	"httpd-foreground"	9 seconds ago	Up 9 seconds	0.0.0.0:80->80/tcp, [::]:80->80/tcp	busy_brahmagupta

```
ubuntu@ip-172-31-45-84:~/app$
```

Building a Docker Image for a Node.js Application

To Begin with the Lab

Summary of the Lab

This lab demonstrated how to install Node.js and npm, create a simple Node.js application with a Dockerfile, build a Docker image, deploy the application in a Docker container, access it through the EC2 public IP, and monitor container logs for verification.

- To install the Node.js, run the command

```
sudo apt update  
sudo apt install -y nodejs npm
```

- Verify the node.js version

```
node -v  
npm -v
```

- Create a directory and create a server.js and Dockerfile inside that directory

```
ubuntu@ip-172-31-45-84:~$ mkdir demoapp  
ubuntu@ip-172-31-45-84:~$ cd demoapp/  
ubuntu@ip-172-31-45-84:~/demoapp$ nano Dockerfile  
ubuntu@ip-172-31-45-84:~/demoapp$ nano server.js  
ubuntu@ip-172-31-45-84:~/demoapp$
```

- Now build the image

```
ubuntu@ip-172-31-45-84:~/demoapp$ sudo docker build -t node-demo .  
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.  
Install the buildx component to build images with BuildKit:  
https://docs.docker.com/go/buildx/  
  
Sending build context to Docker daemon 3.584kB  
Step 1/4 : FROM node  
latest: Pulling from library/node  
aa3e9ef32f73: Pulling fs layer  
a90a67095ead: Pulling fs layer  
0b8a6fde6967: Pulling fs layer  
a47ef1569bfb: Pulling fs layer  
3f59c84a7863: Pulling fs layer  
0252e6abaf0f: Pulling fs layer  
30d0db852850: Pulling fs layer  
a90a67095ead: Download complete  
0b8a6fde6967: Download complete  
4d9ca9dbb25d: Download complete  
5b6d9b49d179: Download complete  
3f59c84a7863: Download complete  
aa3e9ef32f73: Download complete  
a47ef1569bfb: Download complete  
30d0db852850: Download complete  
0252e6abaf0f: Download complete  
aa3e9ef32f73: Pull complete  
3f59c84a7863: Pull complete  
30d0db852850: Pull complete  
0b8a6fde6967: Pull complete  
0252e6abaf0f: Pull complete  
a47ef1569bfb: Pull complete  
a90a67095ead: Pull complete  
Digest: sha256:b46a10d964ad15136ebdf9012142131481caa0697d7a4d4eafe4bbabd818f876  
Status: Downloaded newer image for node:latest  
--> b46a10d964ad
```

- Before running this image stop all the containers
- Then run this one

```
sudo docker run -d -p 80:80 --name node-demo-container node-demo
```

```
ubuntu@ip-172-31-45-84:~/demoapp$ sudo docker images
INFO In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
httpd:latest 393435e1a31 177MB 47.6MB
nginx:latest ec4ed8b5299e 241MB 66MB
node-demo:latest a6dcfec18cf0 1.77GB 442MB
node:latest b46a10d964ad 1.79GB 460MB
static-app:latest 697da8395c47 175MB 45.2MB
static-app:v2 d48b367577a2 175MB 45.2MB
ubuntu:latest b7f48194d4d8 160MB 45.3MB

ubuntu@ip-172-31-45-84:~/demoapp$ sudo docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
f52c13ef739b static-app:v2 "httpd-foreground" 33 minutes ago Exited (0) 2 minutes ago busy_brahmagupta
030e114d0a9b static-app "httpd-foreground" 2 hours ago Exited (0) 2 hours ago nostalgic_nobel
8a6b8279ffdf ubuntu "/bin/bash" 4 hours ago Exited (137) 2 hours ago hopeful_dirac
42b8a28dc825 ubuntu "/bin/bash" 4 hours ago Exited (0) 4 hours ago pedantic_greider
f580899d9b66 ubuntu "/bin/bash" 4 hours ago Exited (0) 4 hours ago gifted_brahmagupta
9791cca36102 ubuntu "/bin/bash" 4 hours ago Exited (0) 4 hours ago youthful_simoussi
47552a7519a2 nginx "/docker-entrypoint..." 4 hours ago Exited (0) 4 hours ago zealous_khayyam
2750e4b5b98f nginx "/docker-entrypoint..." 5 hours ago Exited (0) 4 hours ago practical_greider
7c179fd50eb7 nginx "/docker-entrypoint..." 22 hours ago Exited (137) 21 hours ago wonderful_wiles

ubuntu@ip-172-31-45-84:~/demoapp$ sudo docker run -d -p 80:80 --name node-demo-container node-demo
1f08a6064843d0582a6e2bdda059dd37a9224ald0134aa6820e702b238806f9c
ubuntu@ip-172-31-45-84:~/demoapp$
```

- Test the Server Locally
- In another terminal or browser, visit:
- <http://<EC2-Public-IP>/>



- Viewing Docker Container Logs
- Objective: View logs generated by applications running inside Docker containers.
- Verify the Running Container

```
sudo docker ps
```

- Then run the command

```
sudo docker logs ffd
```

```
ubuntu@ip-172-31-45-84:~/demoapp$ sudo docker run -d -p 80:80 --name node-demo-container node-demo
ffdf29a2f6417dcf32fbd94afa0509024b012f8b57b9e6fb37f4247bf7df0143
ubuntu@ip-172-31-45-84:~/demoapp$ sudo docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
ffdf29a2f641 node-demo "docker-entrypoint.s..." 12 seconds ago Up 11 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp node-demo-container
ubuntu@ip-172-31-45-84:~/demoapp$ sudo docker logs ffd
Server is running on port 80
ubuntu@ip-172-31-45-84:~/demoapp$ sudo docker logs ffd
Server is running on port 80
ubuntu@ip-172-31-45-84:~/demoapp$
```

Building a Docker Image for a PHP Application

To Begin with the Lab

Summary of the Lab

This lab demonstrated how to create a custom Docker image for a PHP web application using the Apache-based PHP image, build and verify the image, manage containers, deploy the application, and access it successfully through the EC2 instance's public IP.

- Objective: Create a custom Docker image for a PHP web application using an Apache-based PHP image.
- Create a Dockerfile
- Create a file named Dockerfile.

nano Dockerfile

- Add the following content:

FROM php:8.2-apache

COPY . /var/www/html

(The instructor uses a PHP image that already includes Apache. The exact PHP version tag may vary.)

- Save the file.

```
ubuntu@ip-172-31-45-84:~/demoapp$ cd ..
ubuntu@ip-172-31-45-84:~$ mkdir PHP
ubuntu@ip-172-31-45-84:~$ cd PHP/
ubuntu@ip-172-31-45-84:~/PHP$ nano Dockerfile
ubuntu@ip-172-31-45-84:~/PHP$
```

- Build the Docker Image

```
sudo docker build -t php-app:v1 .
```

```

ubuntu@ip-172-31-45-84:~/PHP$ sudo docker build -t php-app:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.048kB
Step 1/2 : FROM php:8.2-apache
8.2-apache: Pulling from library/php
4f4fb700ef54: Pulling fs layer
43654fbbfeb1: Pulling fs layer
a4efe98a7f4f: Pulling fs layer
05f27c6ae938: Pulling fs layer
e9571d1330ad: Pulling fs layer
ddd74371c451: Pulling fs layer
d5c4f02e98eb: Pulling fs layer
07c774bd9142: Pulling fs layer
70163b1602d1: Pulling fs layer
ccb8dce9264b: Pulling fs layer
035732560bf3: Pulling fs layer
4b1ebd1b6a41: Pulling fs layer
f3170b4790dd: Pulling fs layer
b206b1362c59: Pulling fs layer
4f4fb700ef54: Already exists
70163b1602d1: Download complete
b206b1362c59: Download complete
05f27c6ae938: Download complete
e9571d1330ad: Download complete
d5c4f02e98eb: Download complete
035732560bf3: Download complete
4b1ebd1b6a41: Download complete
f3170b4790dd: Download complete

```

- Verify the Image

```
sudo docker images
```

```

ubuntu@ip-172-31-45-84:~/PHP$ sudo docker images

IMAGE                ID                DISK USAGE    CONTENT SIZE    EXTRA
httpd:latest         393435ee1a31      177MB         47.6MB
nginx:latest         ec4ed8b5299e      241MB         66MB
node-demo:latest     a6dcfec18cf0      1.77GB        442MB
node:latest          b46a10d964ad      1.79GB        460MB
php-app:v1           05968870281f      707MB         176MB
php:8.2-apache       848bd4a97499      714MB         183MB
static-app:latest    697da8395c47      175MB         45.2MB
static-app:v2        d48b367577a2      175MB         45.2MB
ubuntu:latest        b7f48194d4d8      160MB         45.3MB
ubuntu@ip-172-31-45-84:~/PHP$ █

```

- Stop Any Existing Container Using Port 80

```
sudo docker ps
```

- Stop the container if one is already using port 80.

```
sudo docker stop <container-name>
```

- Run the PHP Application

```
sudo docker run -d -p 80:80 --name php-app-instance php-app:v1
```

- Verify the Container

```
sudo docker ps
```

```

ubuntu@ip-172-31-45-84:~/PHP$ sudo docker images

IMAGE                ID                DISK USAGE    CONTENT SIZE    EXTRA
httpd:latest         393435ee1a31      177MB         47.6MB
nginx:latest         ec4ed8b5299e      241MB         66MB
node-demo:latest     a6dcfec18cf0      1.77GB        442MB
node:latest          b46a10d964ad      1.79GB        460MB
php-app:v1           05968870281f      707MB         176MB
php:8.2-apache       848bd4a97499      714MB         183MB
static-app:latest    697da8395c47      175MB         45.2MB
static-app:v2        d48b367577a2      175MB         45.2MB
ubuntu:latest        b7f48194d4d8      160MB         45.3MB
ubuntu@ip-172-31-45-84:~/PHP$ sudo docker ps

CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                    NAMES
970913b87b54   php-app:v1   "docker-php-entrypoi..."  4 seconds ago   Up 3 seconds   0.0.0.0:80->80/tcp, [::]:80->80/tcp   php-app-instance
ubuntu@ip-172-31-45-84:~/PHP$ █

```

- Access the Application
- Open:
 - `http://<EC2-Public-IP>/index.php`
- The PHP application should be displayed successfully.

Hello from Cloudfreeks!

Current Time: 2026-07-08 10:03:39

PHP Version: 8.2.32